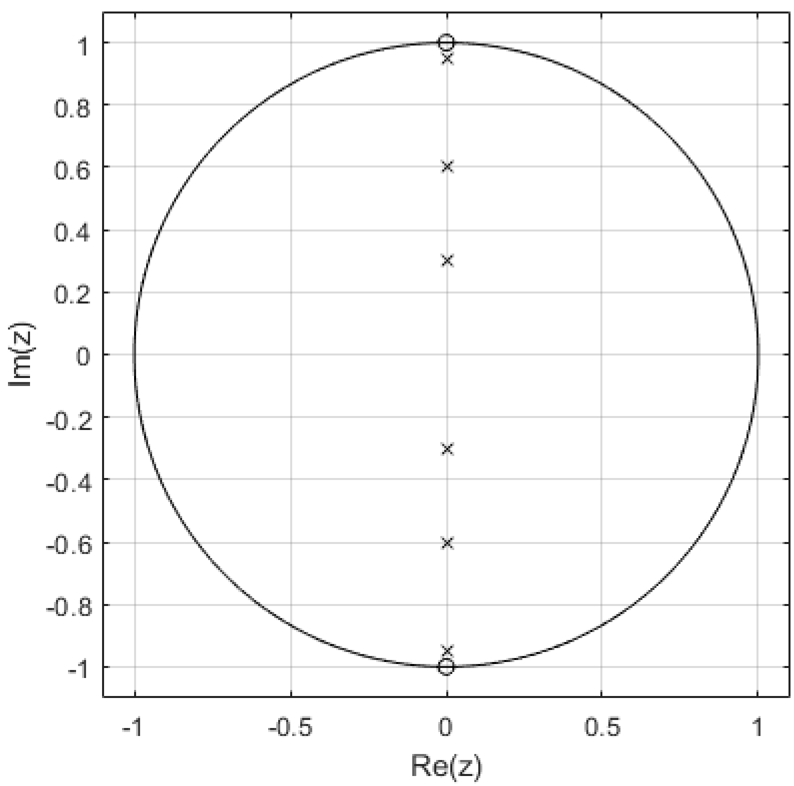
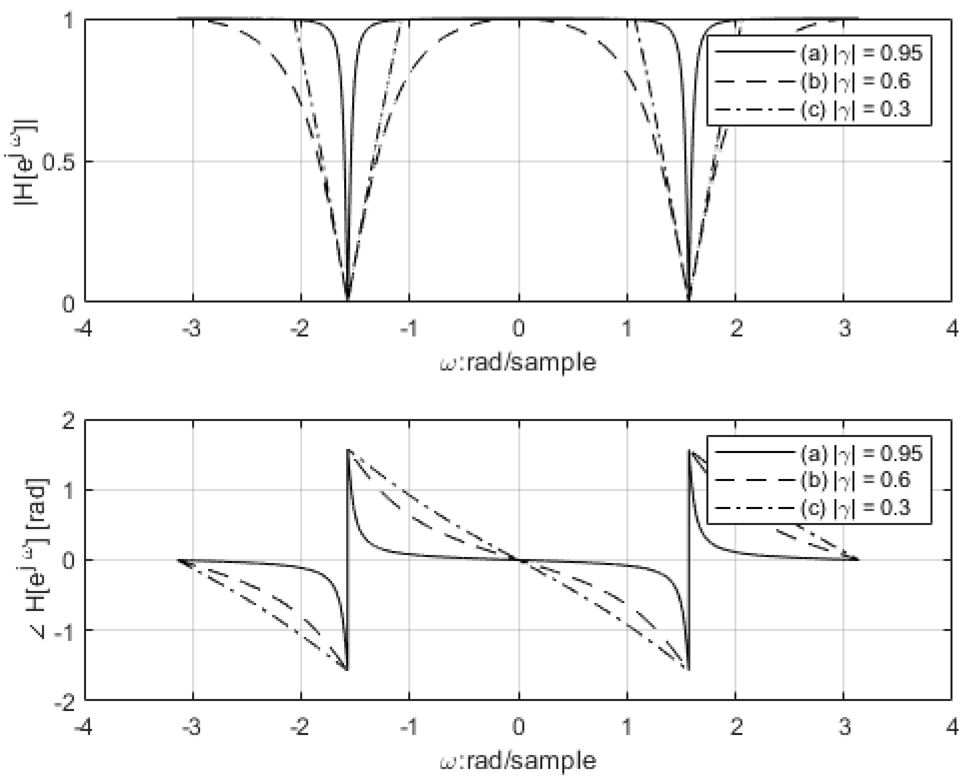
```
% Digital Notch filter frequency response
% Justin Pulley

clear all
close all
clc

a=[0.95 0.6 0.3];
zz = [1j,-1j];
pz = [a*1j,-a*1j];
ucirc = exp(1j*linspace(0,2*pi,201));
plot(real(zz),imag(zz),'ko',real(pz),imag(pz),'kx',...
real(ucirc),imag(ucirc),'k-');
grid on;
xlabel('Re(z)');
ylabel('Im(z)');
axis equal;
axis([-1.1 1.1 -1.1 1.1]);
```




```
omega = linspace(-pi,pi,4097);
a = [0.95 0.6 0.3];
H = zeros(length(a),length(omega));
for m = 1:length(a)
    H(m,:) = freqz([1 0 1],[1 0 a(m)^2],omega);
end
```

```
figure
subplot(2,1,1);
plot(omega,rescale(abs(H(1,:))),'k-',omega,rescale(abs(H(2,:))),'k--',omega,abs(H(3,:)),'k
axis auto;
grid on
xlabel('\omega:rad/sample');
ylabel('|H[e^{j \omega}]|');
legend('(a) |\gamma| = 0.95','(b) |\gamma| = 0.6','(c) |\gamma| = 0.3')
axis ([-4 4 0 1])

subplot(2,1,2);
plot(omega,angle(H(1,:)),'k-',omega,angle(H(2,:)),'k--',omega,angle(H(3,:)),'k-.');
axis auto;
grid on
xlabel('\omega:rad/sample');
ylabel('\angle H[e^{j \omega}] [rad]');
legend('(a) |\gamma| = 0.95','(b) |\gamma| = 0.6','(c) |\gamma| = 0.3')
```



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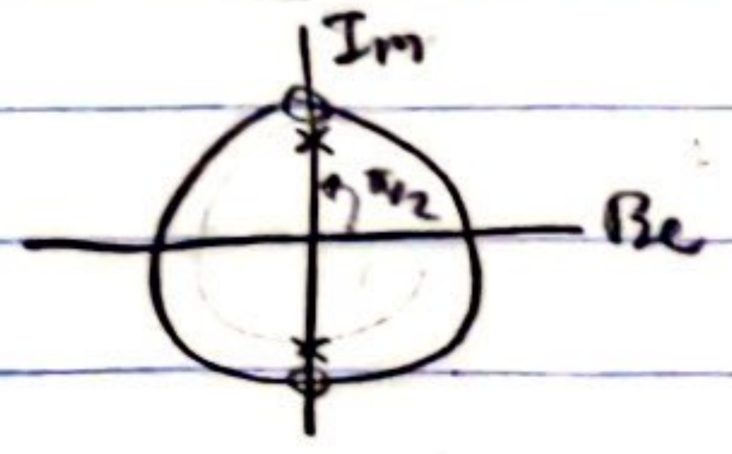
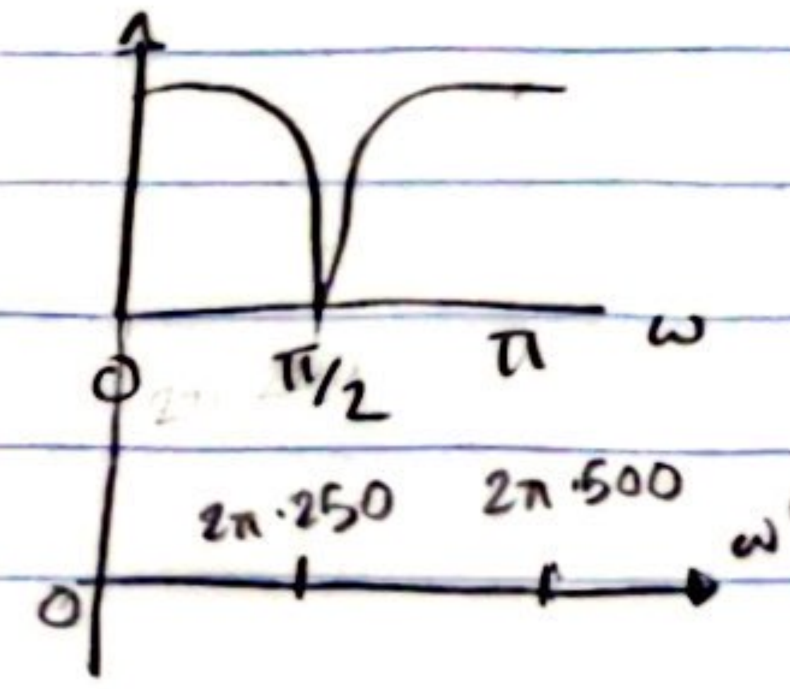
Justin Pulley $\qquad$ 5/3/24

$|H| = 0$ at $\omega' = 250$ Hz, resonance $\omega'_r = 0$ & $2\pi\, 500 \Rightarrow |H|$ maximum

Maximum freq. $\omega'_{highest} = 2\pi \times 500$ rad/s





Solution

since $\omega'_{highest} = 2\pi \cdot 500$ rad/s

$\Rightarrow \omega'_{Nyquist} = 2\pi \cdot 1000$

$\Rightarrow T \le \frac{1}{1000}$ sampling period

~~see |H| = 0 at ω = ω'T = π/2 & −π/2~~

~~|H| max at ω_r = ω_r'T = ... 1000~~

place zeros at $\pi/2$ & $-\pi/2$

place poles near $\pi/2$ & $-\pi/2$ $\Rightarrow a^{+}j$

so: $z_1 = e^{j\pi/2}$, $z_2 = e^{-j\pi/2}$, $\gamma_1 = |\gamma| e^{j\pi/2}$, $\gamma_2 = |\gamma| e^{-j\pi/2}$, $|\gamma| < 1$ (close to 1)

so: $$H(z) = k\,\frac{(z - e^{j\pi/2})(z - e^{-j\pi/2})}{(z - |\gamma| e^{j\pi/2})(z - |\gamma| e^{-j\pi/2})} = \frac{z^2 \cancel{- z e^{-j\pi/2}} \cancel{- z e^{j\pi/2}}^{\,0} - 1}{z^2 \cancel{- z|\gamma| e^{-j\pi/2}} \cancel{- z|\gamma| e^{j\pi/2}}^{\,0} + |\gamma|^2}$$

apparently: $-z|\gamma| e^{-j\pi/4} - z|\gamma| e^{j\pi/4} = -\sqrt{2}|\gamma| z$

$(-e^{-j\pi/4} - e^{j\pi/4}) z|\gamma| \Rightarrow -(2(e^{-j\pi/4} + e^{j\pi/4})) \rightarrow -2\cos(\pi/4)$

so: $(-z e^{-j\pi/2} - z e^{j\pi/2}) \Rightarrow -(e^{-j\pi/2} + e^{j\pi/2}) z|\gamma|\; 2$

$\Rightarrow -2\cos(\pi/2) \Rightarrow 0$

$$\Rightarrow \frac{z^2 + 1}{z^2 + |\gamma|^2} \Rightarrow H(e^{j\omega}) = \frac{e^{j2\omega} + 1}{e^{j2\omega} + |\gamma|^2} = \frac{2}{1 + 0.95^2} = 1.05$$